Rithul Kamesh

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SUMMARY

AI infrastructure and systems engineer building the runtime, storage, and tooling layer beneath LLM and ML workloads. Designed and shipped a multi-tier LLM caching runtime achieving 90%+ token reduction on agent workloads, and an embedded vector database distributed across five language ecosystems. Deep systems background in Rust, Go, C++, and Python spanning execution engines, distributed systems, and developer tooling.

EDUCATION

PES University *Bengaluru, India*
B.Tech, Electronics and Communication Engineering *Aug 2023 – May 2027*

BGS National Public School *Bengaluru, India*
HSSC, SSC *Feb 2023*

EXPERIENCE

H2LooP *On-Site*
AI Developer Advocate Intern *July 2026 – October 2026*

- Built technical demos and example projects showcasing Hydron's AI-powered embedded development platform across firmware generation, debugging, porting, and hardware-aware workflows
- Generated developer-focused technical content and led community initiatives (hackathons, workshops, university outreach) to drive adoption of Hydron among embedded engineers

SELECTED PROJECTS

Continuum *C++, Python*
Unified LLM and ML Execution Runtime

- Designed a unified IR dispatching LLM token generation and tensor computation through a single C++ execution graph, with a five-tier reuse hierarchy — exact memoization, semantic caching, trie prefix KV caching, layer KV warm-start, and memory graph recall — achieving 92.5% token reduction on mixed agent workloads and up to 99% on shared-prefix calls
- Built provider-agnostic backend support (OpenAI, Azure OpenAI, Anthropic, vLLM, libtorch, Apple MLX) with durable, checkpointable execution enabling process-resumable and forkable agent workflows
- Shipped as `continuum-ai` on PyPI with Python/C++ bindings, CI matrix on Linux and macOS, fuzz workflow, and coverage gates; featured on Product Hunt at launch

Sunday *Rust, TypeScript, Tauri, Ollama, SQLite*
Local-First AGI Agent Harness

- Building a privacy-first desktop agent combining voice interaction, screen perception, persistent memory, and computer-use tools in a model-agnostic harness, with local inference via Ollama and durable on-device state
- Engineering an agent architecture around typed actions, approval-gated execution, and auditable tool use, with a plan to evaluate autonomous multi-step task completion against a fixed 50-task benchmark
- Developing an evaluation-driven learning loop to turn successful executions into reusable skills, testing whether orchestration, verification, and memory can improve agent capability independently of model scale

Docproc *Python*
Document Intelligence Engine

- Built a document-to-Markdown conversion engine preserving layout, tables, and semantic structure for LLM ingestion pipelines, with vision-model-based figure and diagram description
- Shipped an evaluation harness with a benchmarking suite measuring downstream reasoning fidelity across document types

RESEARCH EXPERIENCE

Photonics and Quantum Technology Lab, PES University *Bengaluru, India*
Undergraduate Research Assistant *Feb 2025 – Present*

- Modeling photonic integrated circuits using Lindblad/Hamiltonian formalisms for open quantum system simulation under realistic decoherence and noise models
- Developing FDTD-coupled quantum solvers implementing Crank–Nicholson time-stepping with CPML boundaries and NTFF transforms for photonic device characterization
- Investigating state fidelity degradation mechanisms in multi-level quantum systems toward practical photonic qubit implementations

Teaching Assistant, PES University *Bengaluru, India*
Non-Linear Optics and Quantum Technology *Jan 2026 – May 2026*

PUBLICATIONS

B. S. Ullal, V. K. Narasimhaiah, and **R. Kamesh**, "ResNet based deep learning approach for chronic obstructive pulmonary disease prediction using lung sound analysis," *IAES International Journal of Artificial Intelligence (IJ-AI)* (Scopus), vol. 15, no. 2, pp. 1733–1745, Apr. 2026. [ResNet: 97.90% accuracy, 98.72% precision, 0.9937 AUC] doi:10.11591/ijai.v15.i2.pp1733-1745

TECHNICAL SKILLS

Languages: Rust, Go, Python, C/C++, CUDA
AI / ML: PyTorch, Transformers, LangChain, LangGraph, Agent Systems, RAG, LLM Evaluation, HNSW, Vector Databases
Backend: REST, gRPC, FastAPI, Redis, Redis Streams, PostgreSQL, pgvector, BM25, WebSockets
Infrastructure: Docker, Kubernetes, Prometheus, Grafana, OpenTelemetry, CI/CD
Systems: Distributed Systems, SIMD, FDTD, WGPU, OpenMP, Lock-free Concurrency

ACHIEVEMENTS

Hack Night 7.0, Third Place	Oct 2025
Speaker, Snowflake User Meetup Chennai	Jun 2025
Hack Night 6.0, Second Place	Oct 2024